Not All Mathematical Difficulty Is the Same

Likun Xie

> *Some mathematical problems are hard because the solution is difficult to find. Others are hard because we do not yet understand the mathematics needed to solve them. AI's spectacular progress on the first should not cause the public to mistake it for the whole of mathematics.*

People have asked how I feel about AI's rapid progress in mathematics, given concern among some mathematicians and frequent headlines about AI solving difficult problems or mathematicians moving into the AI industry.

I do not share either the anxiety or the hype, nor do I think AI changes what I value deeply in mathematics. I am writing to offer a counterpoint to a public narrative that often uses mathematics as a benchmark for AI progress while presenting mathematical research too narrowly.

A common picture of research is: formulate a problem, try known techniques, and find the clever argument that solves it. AI is becoming very good at parts of this process: suggesting approaches, testing examples, combining methods, and increasingly producing substantial proofs.

But this is only one kind of mathematical difficulty.

Different kinds of mathematical difficulty

Sometimes we understand the mathematics around a problem reasonably well: we know the relevant objects, theories, and techniques, and the main challenge is finding the right sequence of ideas.

$$\text{Problem} \implies \text{try techniques} \implies \text{find the right idea} \implies \text{proof.}$$

This can require great ingenuity, but it also overlaps with what AI is increasingly good at: searching possibilities, combining known methods, and carrying out technical arguments.

At the other end of the spectrum are problems that remain difficult because we do not yet understand the surrounding mathematics deeply enough. We may be missing the right objects, language, connections, or conceptual framework altogether.

$$\text{phenomenon} \implies \text{new viewpoint} \implies \text{new structures} \implies \text{deeper understanding} \implies \text{proof.}$$

Most mathematical problems do not fall neatly into either extreme. They lie somewhere along a spectrum, depending on how much of the relevant conceptual framework is already available and how much new synthesis or theory is required. So I would distinguish three levels, understood as points along this spectrum rather than sharply separated categories:

- *Search within existing techniques.* AI is increasingly strong here.
- *Combine existing concepts in unexpected ways to produce genuinely new proofs.* There is currently much less evidence. Even as AI improves, its effectiveness will likely depend on the scale and conceptual depth of the synthesis required.

- *Recognize that the existing conceptual language is inadequate and create a new mathematical framework that reorganizes part of mathematics.* This is the strongest form of conceptual innovation I am trying to isolate, and current AI appears nowhere close.

The distinction is not simply between problem solving and theory building. Deep theories often emerge from attempts to solve concrete problems, while solving deep problems can itself force us to develop new mathematics.

The direct summand conjecture offers a good example near the conceptual end of the spectrum. Its mixed-characteristic case resisted existing methods for decades before being resolved using perfectoid techniques originating in p-adic geometry. The significance of such a solution lies not merely in finding a proof, but in bringing into the problem a conceptual framework far richer than the original statement. The Weil conjectures and Fermat's Last Theorem offer similar historical examples in which the eventual solution revealed mathematics far larger than the problem itself.

Many problems lie somewhere in between. Suppose, for example, that perfectoid theory already existed but the direct summand conjecture had not yet been solved. Could AI search across existing theories, recognize the relevance of perfectoid methods, and synthesize them into a proof? This does not require inventing perfectoid theory itself, but it still demands substantial insight into which ideas belong together and why. There is much less evidence that AI can reliably handle such problems than tightly specified problems based mainly on established techniques.

This may also help explain why AI first appeared so impressive on exam-style mathematics and other tightly specified tasks: these tend to sit much closer to the first end of the spectrum.

None of this is specific to one area of mathematics, but the balance may differ across fields. Analytic number theory has many highly visible problems involving explicit bounds, asymptotics, or improvements to known results. These are often comparatively well suited to systematic experimentation, computation, and combinations of established techniques, so AI may appear to make faster progress there. Here, technical depth should not be confused with routine work. Many such problems are extremely deep, involving delicate estimates, long chains of reductions, and sophisticated combinations of methods. But when the surrounding conceptual framework is largely in place, this kind of difficulty may still be more amenable to increasingly capable AI than problems requiring deeper conceptual synthesis or new frameworks.

At the same time, analytic number theory also contains questions of great conceptual depth. The twin prime conjecture, for instance, is not simply an invitation to push existing sieve methods a little further; its resolution may require genuinely new insight into the structure and correlations of the primes. Indeed, progress toward it and related problems has repeatedly generated new methods, techniques, and ideas.

These differences may also help explain why mathematicians react so differently to AI. In some research communities, much visible progress consists of technically deep but comparatively well-structured theorem production, so increasingly capable AI could affect those communities more strongly. In others, progress may depend more on conceptual synthesis or new frameworks, where current AI appears much less capable. Perhaps the resulting anxiety reflects real differences in mathematical practice, rather than a single future shared across the subject.

What happens when proofs become cheaper?

We have seen something similar with earlier technological advances. Once computation became cheap, mathematics did not lose its purpose because machines could calculate faster than humans. Mathematical values shifted: large calculations became less interesting in themselves, while understanding the structures behind them became more important.

AI may produce a similar shift at a different level. If certain kinds of technical difficulty become easier to overcome, research may move further toward conceptual depth and more fundamental advances—the direction I find most elegant.

More room for conceptually deep and less visible mathematics

Having interacted with mathematicians across different areas, I have become aware of how differently various kinds of mathematics are perceived, and how some areas that I find especially elegant and deep can be comparatively underappreciated.

The current academic system naturally favors contributions that are easy to make visible, many of which overlap with the kinds of work AI is becoming good at. More exploratory and structural work can be harder to accommodate. In some areas, the conceptual entry cost is itself very high: one may spend years learning a deep theory before even understanding the central questions. Further time may then be needed to find the right framework, connect theories, understand a family of phenomena, or develop language that makes many individual results transparent and potentially helps resolve central problems. The eventual payoff can be enormous, but the intermediate stages are often less visible as conventional research output. Different mathematical communities also assign very different levels of attention and prestige to incremental progress, shaped in part by community size, traditions and shared problems. These differences are reflected in publication cultures as well: some fields produce large numbers of incremental papers, while others tend to publish more selectively and at a slower pace.

This can create a selection effect in academic careers. In areas with high conceptual costs, even very talented mathematicians may leave before their work becomes visible enough to secure opportunities. More visible research cultures, where incremental progress is easier to recognize and reward, may retain a broader range of researchers. If AI makes technically competent, incremental theorem production much cheaper, the prestige attached to such visible output may weaken and potentially reshape these research communities. However, in the short term, AI may have the opposite effect: more papers, more results, and even higher expectations for productivity. That would be an unfortunate transitional equilibrium.

But such an equilibrium becomes harder to sustain if theorem production becomes abundant. Once technically correct, incremental results can be generated cheaply at scale, their sheer number becomes a poor measure of mathematical contribution. This may create more room for work whose value lies not simply in adding another theorem or extending existing methods, but in changing how we understand and connect mathematical ideas.

A larger landscape for individual mathematicians

AI may also change the scale at which an individual mathematician can explore ideas. Today, conceptual breadth is expensive: pursuing a connection between distant subjects may require mastering large amounts of unfamiliar machinery before the idea can even be tested.

A capable mathematical AI could lower these barriers without supplying the central idea itself. It could help navigate literature, explain unfamiliar techniques, test examples, search for obstructions, and handle technical bookkeeping.

If so, an individual researcher could sustain a much broader theoretical program. AI would become not merely a faster tool, but a larger mathematical landscape.

What mathematics do we want?

If the goal were simply to maximize the number of difficult propositions proved, then sufficiently powerful AI would indeed create something close to an existential problem for mathematicians.

But I do not think that is why mathematics is meaningful.

We want to understand why phenomena occur, uncover hidden structures, find the right language, recognize connections between things that initially looked unrelated, and use difficult problems to expose the limits of our current understanding.

Proof is indispensable but proof production is not the whole purpose.

Indeed, for the deepest mathematics, proving and understanding often cannot be separated. A great problem is sometimes valuable precisely because attempting to solve it forces mathematics to acquire concepts that it did not previously have.

If AI makes routine reasoning, technical execution, and some forms of proof search dramatically cheaper, it may free more attention for problems whose difficulty is genuinely conceptual, and give individual mathematicians more room and support to pursue broad and structurally demanding ideas, whose solutions do not merely add another theorem, but change what we understand.

In that sense, AI could make pure mathematics less dominated by the search for the next clever argument and more centered on the deeper reason we solve hard problems in the first place: to discover mathematics that we do not yet know how to see.